

Scenario 1: Generate Monthly Statements

DECLARE

CURSOR GenerateMonthlyStatements IS

SELECT c.CustomerID,

c.Name,

t.TransactionID,

t.TransactionDate,

t.Amount,

t.TransactionType

FROM Customers c

JOIN Accounts a

ON c.CustomerID = a.CustomerID

JOIN Transactions t

ON a.AccountID = t.AccountID

WHERE EXTRACT(MONTH FROM t.TransactionDate) =

EXTRACT(MONTH FROM SYSDATE)

AND EXTRACT(YEAR FROM t.TransactionDate) =

EXTRACT(YEAR FROM SYSDATE);

v_customerid Customers.CustomerID%TYPE;

v_name Customers.Name%TYPE;

v_tid Transactions.TransactionID%TYPE;

v_date Transactions.TransactionDate%TYPE;

v_amount Transactions.Amount%TYPE;

v_type Transactions.TransactionType%TYPE;

BEGIN

OPEN GenerateMonthlyStatements;

LOOP

FETCH GenerateMonthlyStatements

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        INTO v_customerid, v_name, v_tid,
            v_date, v_amount, v_type;

    EXIT WHEN GenerateMonthlyStatements%NOTFOUND;

    DBMS_OUTPUT.PUT_LINE(
        'Customer: ' || v_name ||
        ' | Transaction ID: ' || v_tid ||
        ' | Amount: ' || v_amount ||
        ' | Type: ' || v_type
    );
END LOOP;

CLOSE GenerateMonthlyStatements;
END;
/

```

Scenario 2: Apply Annual Maintenance Fee

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DECLARE

    CURSOR ApplyAnnualFee IS
        SELECT AccountID
        FROM Accounts;

    v_account Accounts.AccountID%TYPE;

BEGIN

    OPEN ApplyAnnualFee;

    LOOP

        FETCH ApplyAnnualFee

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        INTO v_account;

        EXIT WHEN ApplyAnnualFee%NOTFOUND;

        UPDATE Accounts
        SET Balance = Balance - 100,
            LastModified = SYSDATE
        WHERE AccountID = v_account;

        DBMS_OUTPUT.PUT_LINE(
            'Annual fee deducted from Account '
            || v_account
        );

    END LOOP;

    CLOSE ApplyAnnualFee;

    COMMIT;
END;
/

```

Scenario 3: Update Loan Interest Rates

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DECLARE

    CURSOR UpdateLoanInterestRates IS

        SELECT LoanID

        FROM Loans;

    v_loan Loans.LoanID%TYPE;

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BEGIN

  OPEN UpdateLoanInterestRates;

  LOOP

    FETCH UpdateLoanInterestRates
    INTO v_loan;

    EXIT WHEN UpdateLoanInterestRates%NOTFOUND;

    UPDATE Loans
    SET InterestRate = InterestRate + 0.5
    WHERE LoanID = v_loan;

    DBMS_OUTPUT.PUT_LINE(
      'Interest updated for Loan ID '
      || v_loan
    );

  END LOOP;

  CLOSE UpdateLoanInterestRates;

  COMMIT;

END;

/
```